

Are you excited to work on a **high-impact research topic** on **explainability AI for medical report generation**?

The [Chair of Interpretable and Reliable Machine Learning \(EML-Munich\)](#) is looking for YOU!

Project Description

Sparse Autoencoders (SAEs) have shown great promise in detecting concepts related to medical conditions [1]. However, it is not yet well understood how to leverage SAEs in the presence of both known and unknown concepts, for example, from partial labels. Furthermore, their application to downstream tasks such as medical report generation is yet to be explored.

This project aims to use SAEs to discover known and unknown concepts in medical imaging data, with a particular focus on chest x-ray scans. We plan to extract latent activations from specialized models, such as CT-CLIP [2] or MedGemma [3]. We will then train and evaluate different SAE variants on detecting known concepts and subsequently use them for generating medical reports.

The project consists of three main tasks: (1) training SAEs on activations from medical image encoders and evaluating their performance; (2) developing specialized SAE architectures that are better suited for concept extraction; and (3) integrating the SAE features into a medical report generation pipeline, which may be based on Retrieval-Augmented Generation (RAG).

Requirements

- Enrollment in a master's program at the TUM School of CIT with average grade < 1.8
- Completion of relevant lectures such as Introduction to Deep Learning (IN2346), Machine Learning (IN2064) and relevant practical courses.
- Strong motivation to develop state-of-the-art explainability methods with real-world applications.
- Excellent programming skills, with proficiency in Python and experience using deep learning frameworks such as PyTorch.
- (Desirable but not essential) Familiarity with SAEs; prior experience with medical data is a plus.

What We Offer

- Master's thesis
- An exciting topic with significant impact and aim to publish at a top-tier conference
- Working with the friendly & passionate team from [EML-Munich](#), with ample computational resources

Application / Contact

To apply, please send your **CV and transcript** to: explainable.ml.munich@gmail.com with the subject line “**Medical SAEs - EML**”. We'll consider applications until the position is filled, reviewing them starting from **September 5th**.

References

[1] Dasdelen et al. (2025): CytoSAE: Interpretable Cell Embeddings for Hematology. In MICCAI.

[2] Hamamci et al. (2024): A foundation model utilizing chest ct volumes and radiology reports for supervised-level zero-shot detection of abnormalities. In arXiv.

[3] Sellergren et al. (2025): Medgemma technical report. In arXiv.